

MORIOKA, Japan

WMO: 475840

Lat: 39.6986N

Lon: 141.1658E

Elev: 155

StdP: 99.47

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-7.2	-13.0	1.2	-5.2	-11.9	1.4	-4.2	8.5	-0.2	7.5	-0.2	1.9	130	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	31.6	24.0	30.1	23.1	28.6	22.4	24.9	29.6	24.2	28.3	23.6	27.2	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.8	26.8	23.0	18.1	26.4	22.4	17.5	25.9	76.8	29.8	74.2	28.5	71.6	27.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.4	-11.1	34.0	1.7	1.4	-12.3	35.0	-13.3	35.9	-14.3	36.6	-15.5	37.6		
			-11.7	26.1	1.6	0.7	-12.8	26.6	-13.8	27.0	-14.7	27.4	-15.9	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-1.8	-1.0	2.6	8.7	14.7	19.0	23.0	24.1	19.7	12.9	6.3	0.9
	DBStd	9.53	2.55	2.76	3.15	3.61	3.15	2.69	2.76	2.94	3.37	3.15	3.36	2.81
	HDD10.0	1381	366	307	229	66	3	0	0	0	0	9	119	282
	HDD18.3	3179	624	541	487	288	118	24	2	1	23	169	362	540
	CDD10.0	1687	0	0	1	28	149	270	403	438	292	99	7	0
	CDD18.3	443	0	0	0	0	6	44	147	181	65	1	0	0
	CDH23.3	2929	0	0	0	5	78	257	947	1314	325	2	0	0
CDH26.7	827	0	0	0	0	12	44	277	432	62	0	0	0	
Wind	WSAvg	2.9	2.7	2.9	3.2	3.4	3.3	3.1	2.8	2.8	2.6	2.5	2.6	2.6
Precipitation	PrecAvg	1320	54	50	82	91	105	116	193	183	163	112	96	75
	PrecMax	1716	114	126	135	185	193	206	454	436	291	248	185	244
	PrecMin	895	24	20	29	22	38	46	47	52	51	19	24	25
	PrecStd	212	26	24	34	42	39	44	90	102	69	52	35	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	9.2	16.4	23.7	28.4	29.9	33.0	34.2	30.7	23.2	17.3	11.1
		MCWB	3.6	4.9	8.9	12.6	17.0	19.9	24.3	24.9	23.3	17.0	12.5	8.5
	2%	DB	4.4	6.8	12.6	20.5	25.4	27.9	31.2	32.4	28.5	21.2	15.2	8.8
		MCWB	1.7	3.4	7.6	11.9	15.6	19.2	23.7	24.6	22.5	15.9	11.3	6.4
	5%	DB	3.2	5.0	10.4	17.9	23.3	26.2	29.7	30.7	26.6	19.9	13.6	6.9
		MCWB	0.9	2.0	6.2	10.8	15.0	18.7	23.0	23.8	21.2	15.6	10.4	4.6
	10%	DB	2.1	3.6	8.2	15.6	21.2	24.6	28.1	29.1	25.1	18.5	12.1	5.4
		MCWB	0.1	0.9	4.6	9.7	14.1	18.1	22.5	23.3	20.7	14.7	9.5	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	6.0	10.5	15.8	18.7	22.2	25.6	26.0	24.8	19.7	14.1	8.9
		MCDB	5.7	8.0	14.3	19.2	24.6	25.9	30.8	32.0	28.1	20.8	15.9	10.5
	2%	WB	2.4	4.1	8.4	13.3	17.4	21.0	24.6	25.2	23.6	17.8	12.5	6.9
		MCDB	3.6	6.1	11.6	18.1	22.1	24.6	29.3	30.2	26.8	19.8	14.3	8.3
	5%	WB	1.4	2.4	6.7	11.9	16.4	20.0	23.9	24.6	22.7	16.4	11.2	5.0
		MCDB	2.7	4.5	9.8	16.1	20.7	23.9	27.8	29.1	25.8	18.6	13.0	6.6
	10%	WB	0.4	1.2	5.0	10.6	15.3	19.2	23.3	24.0	21.5	15.3	9.8	3.4
		MCDB	1.8	3.1	7.8	14.4	19.4	22.9	26.7	27.9	24.2	17.7	11.7	5.0
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.3	8.4	10.3	10.4	9.0	7.7	7.9	8.3	9.3	8.3	6.2
		MCDBR	6.9	8.8	11.8	14.5	14.8	12.3	10.2	9.9	9.3	10.4	10.4	8.2
		MCWBR	5.2	6.3	7.7	7.6	6.9	5.3	4.1	3.7	4.2	5.9	7.2	6.4
	5% WB	MCDBR	6.4	8.2	10.8	11.1	11.0	9.0	8.3	8.9	7.8	7.9	8.9	7.9
		MCWBR	5.4	6.4	8.0	7.0	5.9	4.4	3.7	3.7	4.0	5.4	7.0	6.6
Clear-Sky Solar Irradiance	taub	0.374	0.421	0.501	0.543	0.556	0.537	0.542	0.518	0.456	0.423	0.403	0.381	
	taud	1.963	1.831	1.771	1.777	1.804	1.922	1.983	2.053	2.185	2.164	2.116	2.061	
	Ebn at Noon	741	759	742	745	746	760	752	758	784	762	713	703	
	Edh at Noon	129	171	202	214	213	190	177	161	133	121	111	110	
All-Sky Solar Radiation	RadAvg	1.72	2.50	3.45	4.42	4.94	4.87	4.31	4.30	3.59	2.84	1.91	1.44	
	RadStd	0.12	0.19	0.24	0.42	0.49	0.37	0.54	0.53	0.29	0.24	0.15	0.13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.35	N/A	N/A	+92	N/A	
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page